Lab 2 (CS201)

1. Radioactive Decay:
   There are two species of radioactive particles A and B. Consider two situations:

   (a) A decays into B and B decays into A. The time constants for the species are τ_A and τ_B respectively. Set up and solve the ODEs by Euler's method for the initial conditions $N_A(0) = 100, N_B(0) = 0$. Can you make an argument for why the behaviour of the solutions depend on τ_A/τ_B? Plot $N_A(t)$ and $N_B(t)$ vs t for the following values of the ration $\tau_A/\tau_B = 0.2, 1.0, 2.2$. Are the results as expected (and why)?

   (b) A decays into B, which decays also (but not into A). For simplicity, take a common time constant $\tau = 1$. Solve the solutions and show that they reach a constant value asymptotically.

   Why does the solution in (a) reach a constant non-zero equilibrium eventually and the solution in (b) does not?

2. Projectile Motion:
   Consider a particle projected at an angle θ with the x-axis from the origin, with a velocity $v = 100m/s$. Taking acceleration due to gravity $g = 10m/s$ and $B_2/m = 10^{-5}$, Taking three different values of θ (of your choice), plot the trajectory $x(t)$ vs $y(t)$ on the same graph using Euler's method. Also perform the same exercise without air resistance (i.e., $B_2 = 0m/s$) and plot $x(t)$ vs $y(t)$. Remark on the differences between the graphs.

   Remark: Try to solve the above problems in as few lines of code as possible. Hint: Although not compulsory, it might be advisable to implement a general function for Euler's method that would be re-usable.